Band-1 Perfect Cuboid Constraint Probe (Collapse 2.30)

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Band-1 Perfect Cuboid / Euler Brick Constraint Probe (Finite + Lit)

Honesty stamp. The existence of a **perfect cuboid** remains **OPEN**. This note does **not** claim a perfect cuboid was found. It does **not** prove nonexistence. 2020–2026 full-nonexistence preprints (Zenodo/arXiv) are treated as **CLAIMS only**, not elevated. Partial fiber results (e.g. torsion-intersection on finitely many master-tuple fibers) are **PARTIAL lit**, not global closure. **No Clay / no ToE. GATE-04** locked out of scope.

Perfect cuboid = OPEN. Finite Euler-brick search $\neq$ existence proof or nonexistence proof.

Abstract

Literature (statement): An *Euler brick* is a rectangular parallelepiped with integer edges a, b, c and integer face diagonals $\sqrt{a^2+b^2}$, $\sqrt{a^2+c^2}$, $\sqrt{b^2+c^2}$. A *perfect cuboid* (perfect Euler brick) additionally has integer space diagonal $\sqrt{a^2+b^2+c^2}$. No perfect cuboid is known; none is proved impossible (Wikipedia *Euler brick*; MathWorld *Euler Brick*; Guy UPNT).

Literature (search bounds): Exhaustive computer searches imply that if a perfect cuboid exists, the odd edge is $> 2.5 \times 10^{13}$, the smallest edge is $> 5 \times 10^{11}$, and the space diagonal is $> 9 \times 10^{15}$ (Matson / Belogourov / Wikipedia summaries). Infinite Euler bricks exist (parametric families); smallest primitive: Halcke $(44, 117, 240)$ (1719).

Box finite check: Brute enumeration of Euler bricks with $1 \le a \le b \le c \le 1000$. Found **10** Euler bricks (**5** primitive); **0** have integer space diagonal. Halcke sanity: face diagonals $(125, 244, 267)$, space$^2 = 73225$ not square — PASS.

Problem (collapse Band-2 item 2.30 → Band-1-style ledger)

1. State Euler brick / perfect cuboid cleanly from the literature.
2. Ledger published search lower bounds and modular necessary conditions.
3. Enumerate small Euler bricks on box; check space diagonal integrality.
4. Sanity-check Halcke $(44, 117, 240)$.
5. Keep Clay, ToE, GATE-04, and "proved nonexistent" off the claim surface.

Literature ledger (web check; not re-proved on box)

Year / item	Result	Status
1719	Halcke brick $(44, 117, 240)$	smallest primitive Euler brick
~1740 / Euler	parametric Euler bricks; perfect cuboid posed	classical

Year / item	Result	Status
Saunderson	infinite parametric family of Euler bricks	classical
Kraitchik (1945)	modular constraints on rational cuboids	published
Korec (1992)	lower bounds; space-diag prime constraints	published
Matson (2015)	odd edge $> 2.5 \times 10^{13}$; smallest $> 5 \times 10^{11}$	computational search
Belogourov / distributed	space diagonal $> 9 \times 10^{15}$	computational search
Wikipedia / MathWorld	existence of perfect cuboid	secondary
	OPEN	
2020–2026	full-nonexistence preprints (various)	**CLAIMS only**
2026	fiber / modular / Mordell–Weil exhaustion drafts	**PARTIAL or CLAIMS** — not elevated

Hygiene. Do **not** elevate Zenodo/arXiv "no perfect cuboid" manuscripts to community closure. Cuboid conjectures (Sharipov irreducibility) are neither proved nor equivalent to the full problem.

Method (box)

1. Pre-scan pairs (a, b) with $a \le b \le C_{\max}$ and $a^2 + b^2$ a perfect square.
2. For each such pair, scan $c \in [b, C_{\max}]$ requiring $a^2 + c^2$ and $b^2 + c^2$ squares.
3. Normalize edges ascending; dedupe; test whether $a^2 + b^2 + c^2$ is square.
4. Spot-check published small primitives (Halcke and Wikipedia list under 1000).
5. Apply published *perfect-cuboid* modular sieve (exactly one odd edge; ≥ 2 edges $\div 3$ with $\ge 1 \div 9$; $\ge 2 \div 4$ with $\ge 1 \div 16$; one each $\div 5, 7, 11, 19$) as a filter on the Euler bricks found — survivors still need integer space diagonal.
6. Label: **finite search** — not a bound at 10^{11}, not a nonexistence proof.

Results

Literature status

Quantity	Value
Perfect cuboid existence (community)	**OPEN**
Smallest known Euler brick	$(44, 117, 240)$ (Halcke)
Lit lower bound (smallest edge)	$> 5 \times 10^{11}$
Lit lower bound (odd edge)	$> 2.5 \times 10^{13}$
Lit lower bound (space diagonal)	$> 9 \times 10^{15}$
Full nonexistence preprints elevated?	**No**

Box window (tonight)

Quantity	Value
$C_{\max}$	1000
Euler bricks found	**10**
Primitive among them	**5**

Quantity	Value
Perfect cuboids found	**0**
Perfect-cuboid modular-sieve survivors	**0**
Halcke found + face diags $(125, 244, 267)$	**PASS**
Halcke space^2 = 73225 not square	**PASS**
Wall (box)	≈ 0.13 s
Label	finite search $\neq$ proof; $\neq$ world search depth

All Euler bricks with max edge ≤ 1000 (box)

Edges (a, b, c)	Face diagonals	Space^2	Primitive	Perfect?
$(44, 117, 240)$	$125, 244, 267$	73225	yes	**no**
$(240, 252, 275)$	$348, 365, 373$	196729	yes	**no**
$(88, 234, 480)$	$250, 488, 534$	292900	no (2×Halcke)	**no**
$(480, 504, 550)$	$696, 730, 746$	786916	no	**no**
$(140, 480, 693)$	$500, 707, 843$	730249	yes	**no**
$(85, 132, 720)$	$157, 725, 732$	543049	yes	**no**
$(132, 351, 720)$	$375, 732, 801$	659025	no	**no**
$(160, 231, 792)$	$281, 808, 825$	706225	yes	**no**
$(720, 756, 825)$	$1044, 1095, 1119$	1770561	no	**no**
$(176, 468, 960)$	$500, 976, 1068$	1171600	no (4×Halcke)	**no**

Matches the five Wikipedia primitives with all edges < 1000: $(44, 117, 240)$, $(240, 252, 275)$, $(85, 132, 720)$, $(140, 480, 693)$, $(160, 231, 792)$.

Near-misses (space diagonal closest to integer by square gap)

Closest gaps among the 10: $(85, 132, 720)$ gap 120; $(480, 504, 550)$ gap 147; Halcke gap 216. None zero.

Why Band-1 vs conjecture

This probe only certifies a **tiny** exhaustive window ($c \leq 1000$) consistent with the open status and with published search floors orders of magnitude larger. Finding Euler bricks is easy; forcing the seventh length (space diagonal) to be integer remains the open Diophantine obstruction. **No claim** that the modular sieve or fiber partials close the problem.

Verdict

Claim	Result
Found a perfect cuboid on box?	**NO** (0 of 10 Euler bricks)
Proved nonexistence on box?	**NO**
Problem status	**OPEN**
Halcke sanity	**PASS**
Clay / ToE / GATE-04	**out of scope / LOCKED**

References (secondary)

1. Wikipedia, *Euler brick* (perfect cuboid section; search bounds; Halcke).
2. Weisstein, Eric W. "Euler Brick." *MathWorld*.
3. R. K. Guy, *Unsolved Problems in Number Theory* (rational cuboid).

4. R. D. Matson, “Results of a Computer Search for a Perfect Cuboid” (2015).
5. M. Kraitchik, “On certain rational cuboids,” *Scripta Mathematica* 11 (1945).
6. I. Korec, “Lower bounds for Perfect Rational Cuboids,” *Math. Slovaca* 42 (1992).

ZoraASI Band-1 dash · collapse 2.30 · 2026-09-11 · GATE-04 locked · no Clay/ToE stamp · perfect cuboid OPEN · 10 Euler bricks / 0 perfect ($c \leq 1000$).